

Statistical and Computational Reproduction of Part III: Unconditional ODTE Trading Rules

M.A. in Statistics - Graduating Project Practicum

Data Analysis, Statistical Methodology, Computational Reproduction and Limitations

Based on Vilkov (2026), “ODTE Trading Rules”

August 2026

1. Data and Empirical Analysis

The empirical analysis is based on the processed research files supplied with the replication package: `data_opt.parquet`, `data_structures.parquet`, `future_moments_SPX.parquet`, `vix.parquet`, and `ALL_eod.csv`. The option-level data contain standardized moneyness, option prices, intrinsic and time value, expiration payoffs, holding-period returns, implied volatility, Greeks, and liquidity measures. The strategy-level data contain the multi-leg positions used in the unconditional trading-rule analysis. The remaining files provide variance- and volatility-related quantities and market series used to characterize intraday compensation and temporal variation. The analysis therefore begins from the authors' processed research sample rather than from raw Cboe quote records.

This distinction determines the scope of the reproduction. Quote filtering, interpolation to the standardized moneyness grid, and some upstream variable construction are inherited from the supplied files. The present project reproduces the downstream statistical analysis of Part III conditional on those inputs. It does not claim to independently reconstruct the raw quote-cleaning process. All subsequent estimates are generated from the supplied data and computational materials without replacing reproduced values by values reported in the paper.

1.1 Observational structure and response variables

The trading day is the principal repeated observational unit. Within each day, several moneyness configurations and strategy families are observed. Let d index trading days, c index a moneyness configuration, s index a strategy family, and j index an option leg. The main response variable is the holding-period profit and loss normalized by the level of the S&P 500 index at entry. Denote this response by $Y_{\{dcs\}}$. This notation makes explicit that configurations observed on the same day share the same underlying market realization and should not automatically be regarded as independent observations.

For an option with strike K and underlying level S_t at the entry time t , moneyness is defined as $m=K/S_t$. The expiration payoffs of calls and puts are $(S_T-K)^{+}$ and $(K-S_T)^{+}$, respectively. If V_t denotes the option entry price, the underlying-normalized option PNL is

$$Y_d = (\text{Payoff}_T - V_t) / S_t \quad (1)$$

and is multiplied by 100 when reported as a percentage of the underlying. This normalization is central to the empirical design. Percentage option returns can become extremely large when the entry premium is close to zero; normalizing PNL by the underlying provides a more stable scale for comparisons across strikes and dates.

1.2 Exploratory distributional evidence

The exploratory analysis is intentionally distributional. For each entry time, moneyness level, or strategy configuration, the empirical mean is reported together with the median, quartiles, selected tail quantiles, dispersion, and skewness. This choice is motivated by the strong asymmetry visible in ODTE outcomes. In such data, the mean alone may be strongly affected by a small number of extreme observations and may not represent the typical trading day.

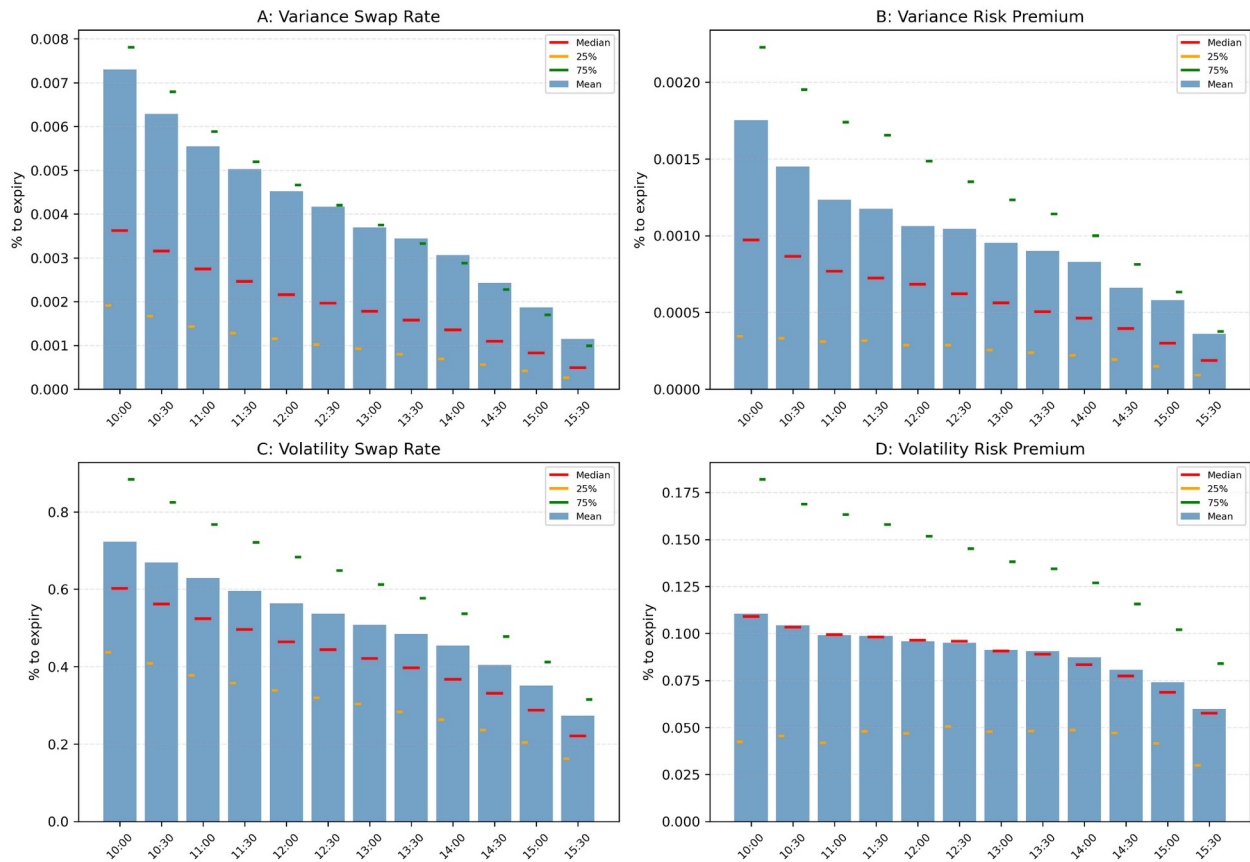


Figure 1. Reproduced intraday variance and volatility compensation.

Figure 1 summarizes the intraday empirical distributions of the variance swap rate, variance risk premium, volatility swap rate, and volatility risk premium from 10:00 to 15:30 ET. All four measures decline as expiration approaches. The mean generally exceeds the median and the upper quartile is separated from the center of the distribution, indicating positive asymmetry. The figure is interpreted descriptively: it characterizes how same-day compensation changes as the remaining horizon contracts, rather than providing a predictive trading rule.

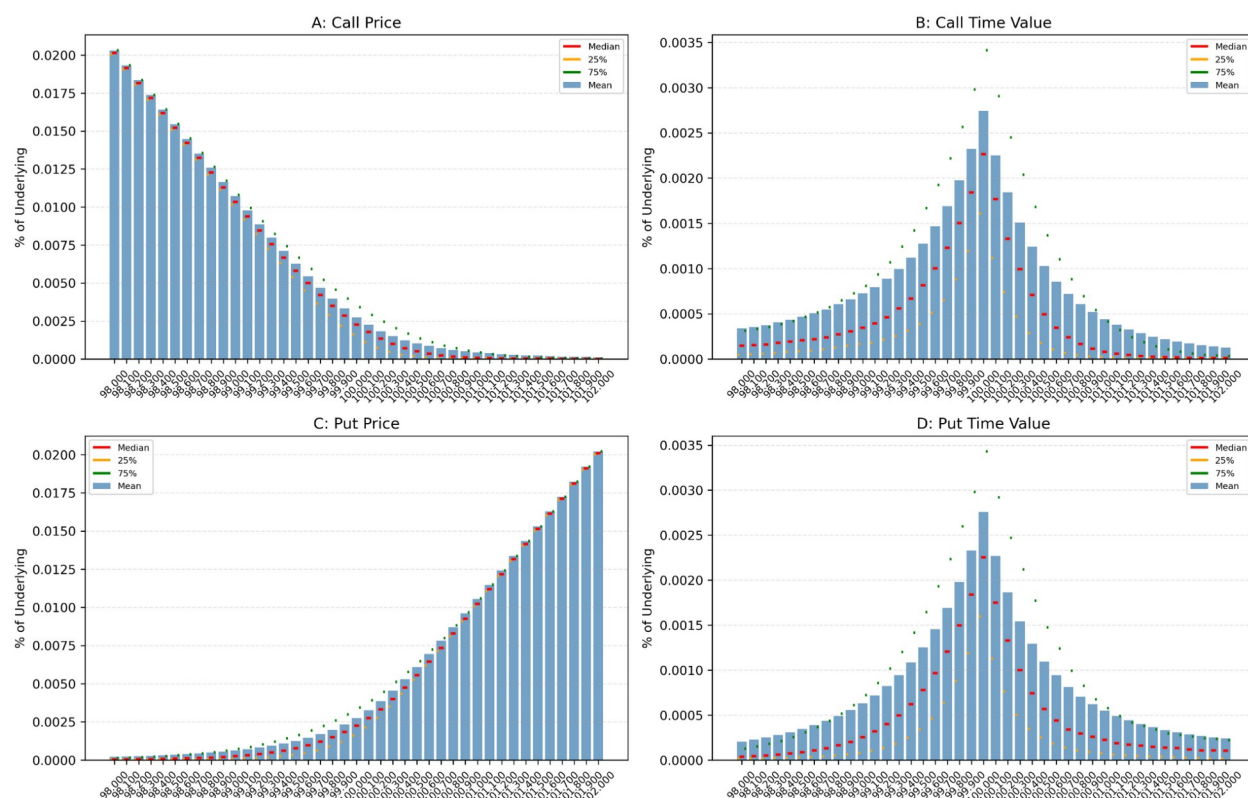


Figure 2. Reproduced call and put prices and time value across moneyness.

The option-price cross-sections in Figure 2 exhibit the expected monotone relation with moneyness: call prices decrease as strike relative to spot increases, whereas put prices increase. Time value is concentrated around the at-the-money region and decays rapidly away from it. For ODTE contracts this concentration is particularly pronounced because only a few hours remain for the terminal payoff to differ from intrinsic value.

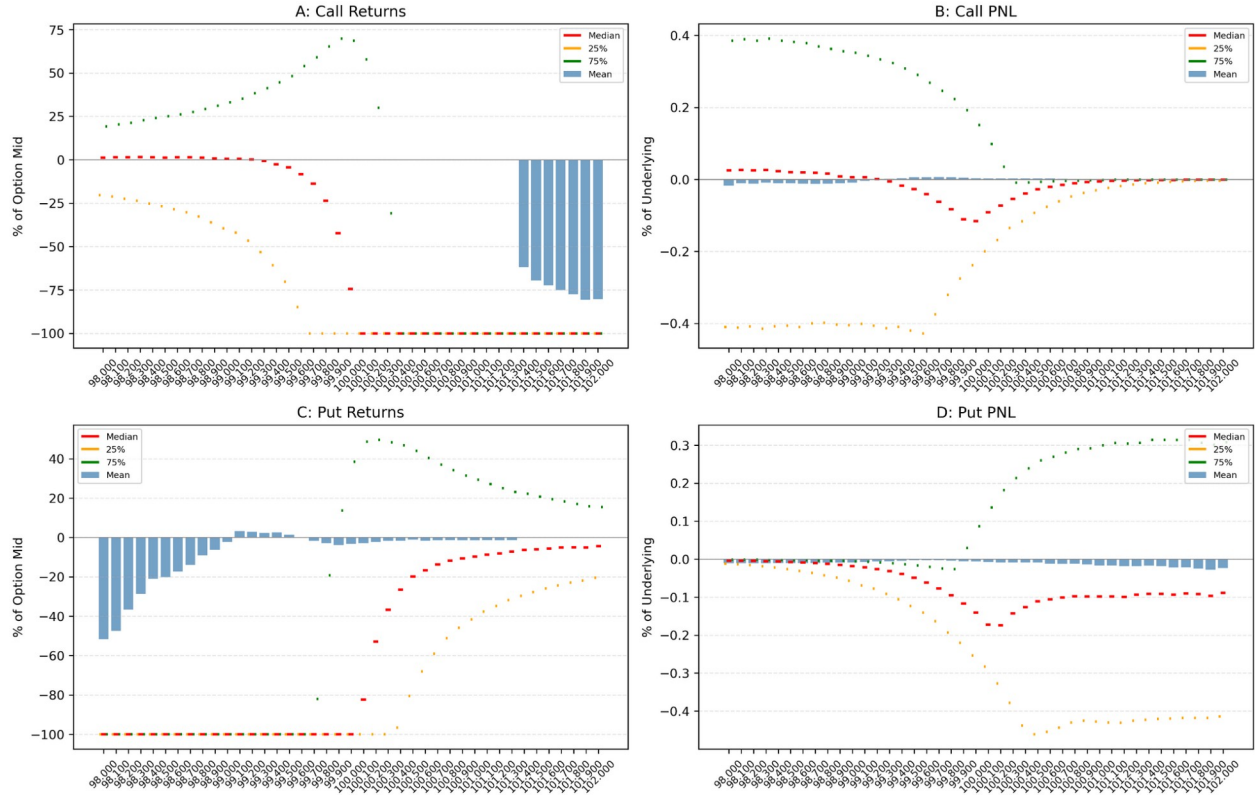


Figure 3. Reproduced individual option returns and underlying-normalized PNL.

Figure 3 highlights a statistical distinction that is important for the remainder of the project. Percentage option returns exhibit extreme values where option premiums are small, while underlying-normalized PNL is considerably more comparable across strikes. The difference is not merely cosmetic: it changes the scale on which cross-sectional dispersion and tail behavior are evaluated. Consequently, the strategy analysis uses underlying-normalized PNL as the principal outcome.

2. Statistical and Computational Methodology

The statistical objective is to estimate and compare unconditional distributions of ODTE strategy outcomes while preserving heterogeneity across moneyness configurations and allowing for dependence within trading days. The computational implementation follows a reproducible sequence from input data to variable construction, grouping, estimation, and graphical or tabular output. The statistical estimands are defined before their empirical estimators so that each reported quantity has a clear interpretation.

2.1 Distributional estimands and empirical estimators

For a fixed strategy s and configuration c , let $Y_1, \dots, Y_n$ denote the observed daily PNL values. The primary location estimand is the unconditional mean $\mu_{\{sc\}} = E(Y_{\{dcs\}})$, estimated by the sample mean

$$\hat{\mu}_{\{sc\}} = n_{\{sc\}}^{-1} \sum_{d=1}^{n_{\{sc\}}} Y_{\{dcs\}} \quad (2)$$

with dispersion estimated by the Bessel-corrected sample variance

$$\hat{\sigma}_{\{sc\}}^2 = (n_{\{sc\}} - 1)^{-1} \sum_{d=1}^{n_{\{sc\}}} (Y_{\{dcs\}} - \hat{\mu}_{\{sc\}})^2 \quad (3)$$

Because the empirical distributions are asymmetric and heavy-tailed, the analysis does not impose a Gaussian model on the static PNL distributions. Instead, quantiles are estimated from the empirical distribution function

$$\hat{F}_{\{sc\}}(y) = n_{\{sc\}}^{-1} \sum_{d=1}^{n_{\{sc\}}} 1\{Y_{\{dcs\}} \leq y\} \quad (4)$$

with empirical p-quantile $\hat{q}_{\{p,sc\}} = \inf\{y: \hat{F}_{\{sc\}}(y) \geq p\}$. This construction provides the 1st, 25th, 50th, 75th, and 99th percentiles reported in the static-performance table. Sample skewness is included to characterize asymmetry, while the annualized Sharpe ratio is reported as a conventional descriptive signal-to-noise measure,

$$SR_{\{sc\}} = \sqrt{252} \cdot \hat{\mu}_{\{sc\}} / \hat{\sigma}_{\{sc\}} \quad (5)$$

but it is not treated as a complete measure of risk. The Sharpe ratio depends only on the first two moments and can therefore obscure the asymmetric tails that are economically important for short-horizon option strategies.

2.2 Construction of multi-leg strategy outcomes

The seven strategy families are strangle/straddle, iron butterfly/condor, risk reversal, bull call spread, call ratio spread, bear put spread, and put ratio spread. For strategy s with J_s option legs, the daily normalized strategy PNL is formed as the signed sum of the leg-level outcomes,

$$Y_{\{dcs\}} = \sum_{j=1}^{J_s} q_{\{js\}} Y_{\{dcjs\}} \quad (6)$$

where $q_{\{js\}} > 0$ denotes a long position and $q_{\{js\}} < 0$ a short position; ratio spreads use absolute quantities larger than one where required. Each supplied moneyness combination is retained as a separate empirical specification. This is statistically preferable to selecting a single representative strike combination because it makes cross-configuration heterogeneity observable rather than absorbing it into an arbitrary average.

2.3 Temporal variation and the 2022 regime specification

Temporal stability is first examined nonparametrically using a 63-observation rolling mean. For an ordered daily series Y_t , the local estimator at date t is

$$\hat{\mu}_t^{(63)} = 63^{-1} \sum_{i=0}^{62} Y_{\{t-i\}} \quad (7)$$

which is approximately a quarter of trading observations. Because adjacent windows share 62 of 63 observations, these rolling estimates are strongly dependent and are used only as descriptive evidence of local variation, not as a sequence of independent hypothesis tests.

Formal inference around the 2022 expansion of daily SPXW expirations follows a pooled fixed-effects specification. Observations through 14 April 2022 form the pre-period, observations from 11 May 2022 form the post-period, and the transition interval is excluded. For each strategy family, the model is

$$Y_{\{dc\}} = \alpha_c + \beta \text{Post}_d + \varepsilon_{\{dc\}} \quad (8)$$

where α_c is a configuration-specific fixed effect and Post_d is an indicator for the post-expansion period. The parameter β is the average change in conditional mean PNL after the regime change, holding configuration fixed. The inferential problem is

$$H_0: \beta = 0 \quad \text{versus} \quad H_1: \beta \neq 0 \quad (9)$$

Standard errors are clustered by trading date. This adjustment is required because multiple configurations observed on the same day are exposed to the same SPX realization and common market conditions. The reported p-value therefore concerns the mean-shift parameter β under model (8). A failure to reject H_0 is not interpreted as evidence that the entire pre- and post-period distributions are equal.

2.4 Implementability and tail-risk estimation

The baseline outcomes are marked at option mid quotes. To approximate implementable performance, the response variable is transformed by subtracting execution frictions. The first adjustment charges one-half of the observed bid-ask spread to each strategy leg. A further 0.5 basis-point slippage-and-fee charge, expressed relative to the underlying, is then deducted. Let Y_d^{mid} denote daily mid-price PNL and C_d the total estimated execution cost. Net PNL is

$$Y_d^{\text{net}} = Y_d^{\text{mid}} - C_d \quad (10)$$

The same descriptive estimators are then applied to the net series. This design allows the effect of transaction costs to be evaluated as a direct change in the empirical response distribution rather than as a separate qualitative judgment.

Tail risk is estimated nonparametrically from the implementable net-PNL series. Let $\hat{q}_\alpha$ be the empirical α -quantile. Expected shortfall at level $\alpha=0.01$ is estimated by

$$ES_\alpha = - [\sum_d Y_d^{\text{net}} 1\{Y_d^{\text{net}} \leq \hat{q}_\alpha\}] / [\sum_d 1\{Y_d^{\text{net}} \leq \hat{q}_\alpha\}] \quad (11)$$

where the sign convention reports the magnitude of adverse tail losses as a positive number. If $C_t = \sum_{d \leq t} Y_d^{\text{net}}$ denotes cumulative PNL and $M_t = \max_{j \leq t} C_j$ the running peak, maximum drawdown is

$$MD\hat{D} = \max_t (M_t - C_t) \quad (12)$$

The analysis also reports the worst one-day outcome, the worst five-day cumulative outcome, and the empirical loss probability $n^{-1} \sum_d 1\{Y_d^{\text{net}} < 0\}$. These quantities are included because the combination of asymmetry and heavy tails makes variance-based risk summaries insufficient.

3. Computational Reproduction and Statistical Results

The reproduction is evaluated at two levels. First, the generated figures and tables are examined for qualitative agreement with the empirical patterns reported in Part III. Second, reproduced numerical estimates are compared with the paper where corresponding benchmark values are available. Numerical discrepancies are retained and discussed rather than calibrated away.

3.1 Static performance across strategies and moneyness

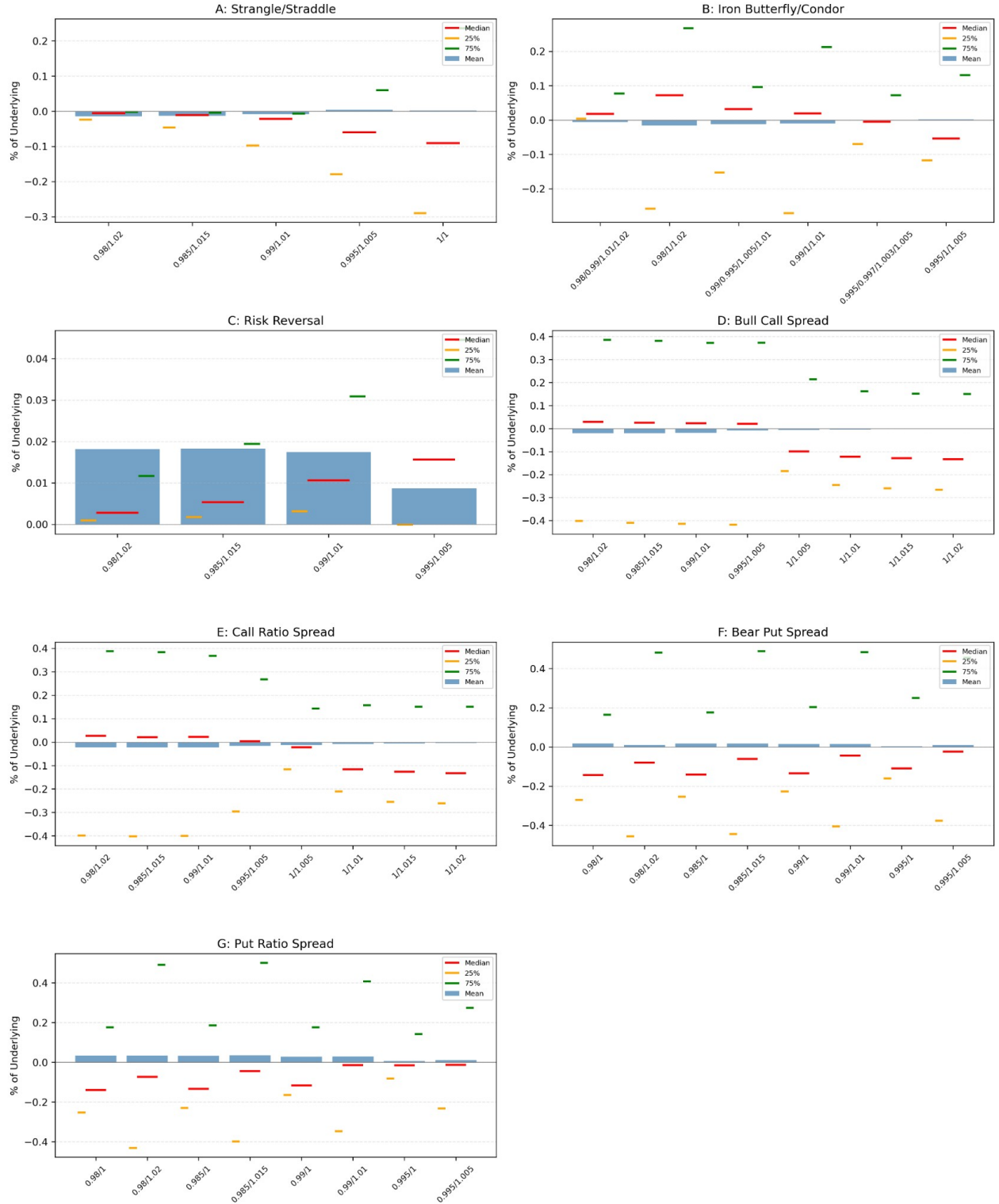


Figure 4. Reproduced static PNL distributions for the seven strategy families.

Strategy	Moneyess	Count	Mean	Volatility	Min	1%	25%	50%	75%	99%	Max	Skew	SR, p.a.
Strangle/ Straddle	1/1	952	0.002	0.541	-2.688	-1.024	-0.290	-0.090	0.236	1.813	2.959	0.952	0.056
Strangle/ Straddle	0.995/1.005	953	0.004	0.461	-2.703	-0.853	-0.179	-0.060	0.060	1.710	2.903	1.361	0.141
Strangle/ Straddle	0.99/1.01	953	-0.008	0.357	-2.753	-0.778	-0.097	-0.022	-0.007	1.485	2.731	1.720	-0.368
Iron Butterfly/Con dor	0.995/1/1.005	952	0.002	0.161	-0.500	-0.254	-0.117	-0.054	0.131	0.354	0.433	0.494	0.217
Iron Butterfly/Con dor	0.99/1/1.01	952	-0.010	0.304	-1.000	-0.654	-0.271	0.019	0.213	0.591	0.835	-0.104	-0.532
Iron Butterfly/Con dor	0.995/0.997/1. 003/1.005	953	-0.001	0.084	-0.202	-0.153	-0.070	-0.005	0.072	0.155	0.181	0.051	-0.171
Iron Butterfly/Con dor	0.99/0.995/1.0 05/1.01	953	-0.012	0.179	-0.500	-0.439	-0.153	0.032	0.096	0.300	0.402	-0.582	-1.097
Risk Reversal	0.995/1.005	953	0.009	0.621	-3.481	-1.848	-0.000	0.016	0.045	2.150	5.358	1.021	0.223
Risk Reversal	0.99/1.01	953	0.017	0.436	-2.999	-1.335	0.003	0.011	0.031	1.662	4.917	2.361	0.634
Bull Call Spread	1/1.005	953	-0.006	0.204	-0.425	-0.241	-0.185	-0.099	0.214	0.355	0.500	0.511	-0.470
Bull Call Spread	1/1.01	953	-0.004	0.326	-0.509	-0.422	-0.245	-0.122	0.162	0.752	1.000	0.920	-0.205
Bull Call Spread	0.995/1.005	953	-0.009	0.379	-0.704	-0.560	-0.418	0.021	0.373	0.506	1.000	-0.108	-0.373
Bull Call Spread	0.99/1.01	953	-0.019	0.602	-1.121	-1.064	-0.415	0.024	0.373	0.999	2.000	-0.108	-0.491

Table 1A. Reproduced static strategy performance: configurations 1-13.

Strategy	Moneyness	Count	Mean	Volatility	Min	1%	25%	50%	75%	99%	Max	Skew	SR, p.a.
Call Ratio Spread	1/1.005	953	-0.012	0.348	-3.552	-1.206	-0.116	-0.022	0.143	0.568	1.798	-3.681	-0.567
Call Ratio Spread	1/1.01	953	-0.009	0.340	-3.050	-0.506	-0.211	-0.115	0.157	0.739	1.261	-1.162	-0.410
Call Ratio Spread	0.995/1.005	953	-0.015	0.408	-3.342	-0.996	-0.295	0.004	0.268	0.745	1.525	-1.641	-0.596
Call Ratio Spread	0.99/1.01	953	-0.023	0.565	-2.649	-1.026	-0.400	0.022	0.368	1.044	1.640	-0.303	-0.652
Bear Put Spread	0.995/1	952	0.004	0.207	-0.268	-0.216	-0.160	-0.108	0.250	0.385	0.449	0.720	0.288
Bear Put Spread	0.99/1	952	0.014	0.345	-0.456	-0.384	-0.228	-0.135	0.203	0.814	0.886	1.061	0.659
Bear Put Spread	0.995/1.005	905	0.009	0.383	-0.576	-0.505	-0.376	-0.023	0.455	0.562	0.768	0.129	0.387
Bear Put Spread	0.99/1.01	784	0.014	0.626	-1.027	-1.003	-0.406	-0.044	0.483	1.070	1.121	0.154	0.357
Put Ratio Spread	0.995/1	952	0.006	0.307	-2.220	-1.235	-0.082	-0.015	0.142	0.578	1.428	-2.159	0.313
Put Ratio Spread	0.99/1	952	0.027	0.320	-1.523	-0.493	-0.165	-0.117	0.176	0.855	1.985	0.955	1.349
Put Ratio Spread	0.995/1.005	905	0.011	0.374	-2.003	-1.012	-0.232	-0.014	0.274	0.806	1.679	-0.412	0.459
Put Ratio Spread	0.99/1.01	784	0.028	0.569	-1.157	-0.971	-0.347	-0.015	0.407	1.250	2.459	0.308	0.791

Table 1B. Reproduced static strategy performance: configurations 14-25. PNL is normalized by the underlying; SR p.a. is annualized by $\sqrt{252}$.

Table 1 retains all 25 reproduced strategy-moneyness configurations. The table illustrates why conclusions should be configuration-specific. For example, the risk reversal has positive reproduced means for both displayed configurations, whereas the sign of the strangle/straddle mean changes across moneyness. The put ratio spread at 0.99/1 has mean PNL 0.0272 and annualized Sharpe ratio 1.349, but its lower quantiles remain negative. Thus a positive average does not imply uniformly favorable outcomes, and the empirical distribution must be considered jointly with the mean.

3.2 Time variation and structural-break inference

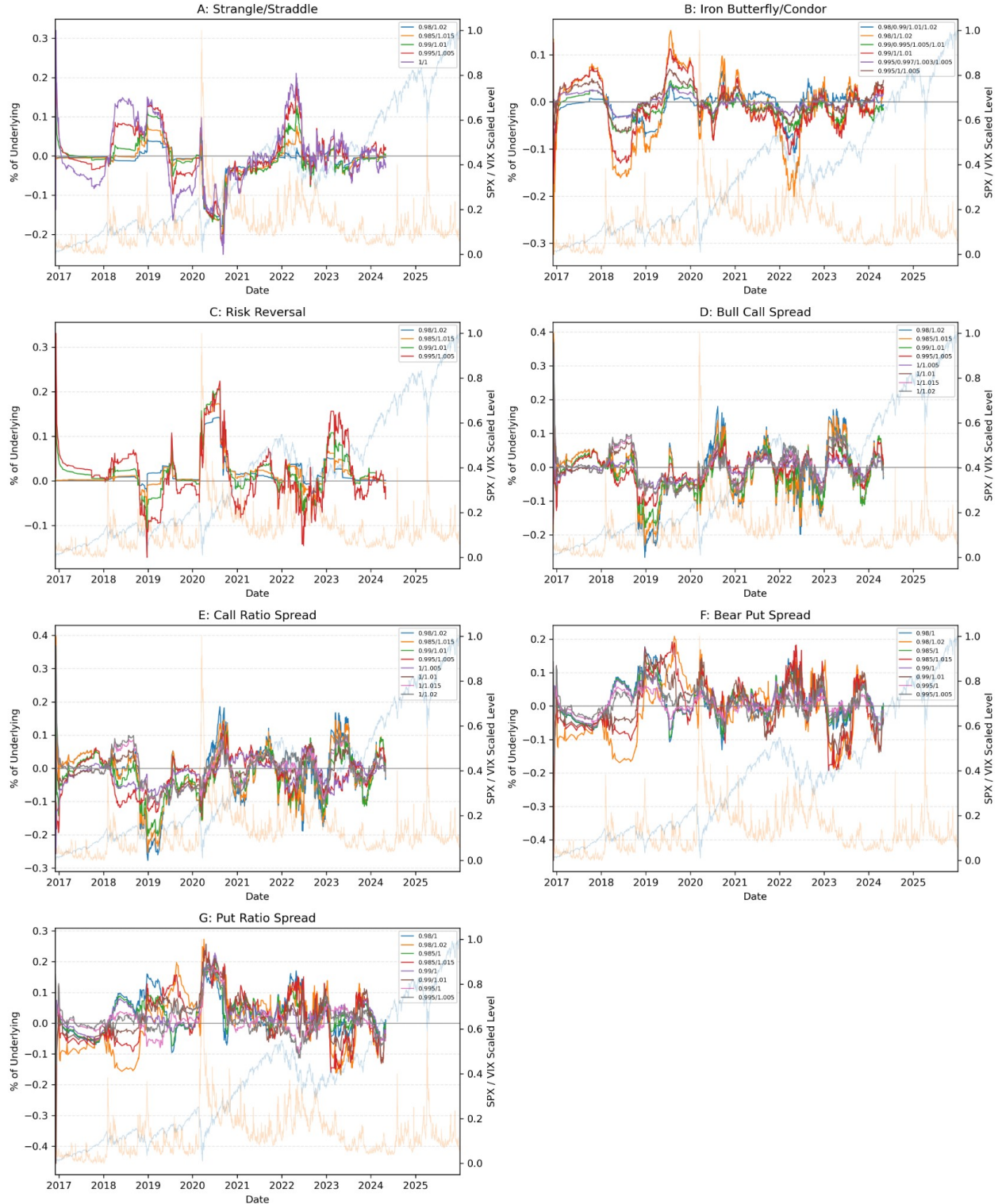


Figure 5. Reproduced 63-observation rolling PNL with scaled SPX/VIX context.

The rolling estimates in Figure 5 change sign repeatedly and display pronounced variation around stressed market periods. This instability indicates that a full-sample unconditional mean can conceal important temporal heterogeneity. The figure therefore provides descriptive motivation for the formal regime analysis rather than independent evidence of a discrete break.

Strategy	Pre Mean	Post Mean	Delta	t	p-value	Vol Ratio	N Pre	N Post
Strangle/ Straddle	-0.008	-0.010	-0.001	-0.053	0.958	0.914	2952	1745
Iron Butterfly/Cond or	-0.009	-0.001	0.008	0.590	0.555	1.067	3541	2094
Risk Reversal	0.024	0.005	-0.019	-0.816	0.414	0.829	2362	1396
Bull Call Spread	-0.012	-0.002	0.010	0.313	0.755	1.060	4724	2792
Call Ratio Spread	-0.019	0.000	0.019	0.690	0.490	1.037	4724	2792
Bear Put Spread	0.019	-0.002	-0.021	-0.652	0.515	1.052	4067	2660
Put Ratio Spread	0.038	0.006	-0.033	-1.220	0.222	1.034	4067	2660

Table 2. Reproduced 2022 structural-break estimates.

For all seven strategy families, the reproduced p-values exceed 0.05. At the conventional 5% significance level, the null hypothesis $H_0:\beta=0$ is therefore not rejected. The correct statistical conclusion is that the supplied sample does not provide sufficient evidence of a change in conditional mean PNL under specification (8). It would be stronger than the evidence permits to conclude that there was no structural change. Volatility ratios, rolling behavior, and tail characteristics may change even when the mean-shift coefficient is not statistically distinguishable from zero.

3.3 Implementability after execution costs

Strategy	Mean Mid	Mean B/A	Mean B/A+0.5bp	SR Mid	SR B/A+0.5bp	Turnover (bps)	SR/ Turnover	ES1%	Mean/ ES1%	Obs
Strangle/ Straddle	-0.006	-0.006	-0.011	-0.271	-0.505	0.241	-2.097	1.423	-0.008	953
Iron Butterfly/Co ndor	-0.007	-0.008	-0.013	-0.556	-0.956	0.310	-3.084	0.579	-0.022	953
Risk Reversal	0.016	0.016	0.011	0.654	0.442	0.027	16.569	1.601	0.007	953
Bull Call Spread	-0.010	-0.010	-0.015	-0.346	-0.528	0.791	-0.668	0.904	-0.017	953
Call Ratio Spread	-0.014	-0.014	-0.019	-0.550	-0.759	0.749	-1.013	1.209	-0.016	953
Bear Put Spread	0.014	0.014	0.009	0.480	0.296	0.629	0.470	0.827	0.010	952
Put Ratio Spread	0.025	0.025	0.020	1.064	0.838	0.574	1.460	0.710	0.028	952

Table 3. Reproduced implementability diagnostics.

Table 3 demonstrates that execution costs materially alter the empirical performance distribution. After the bid-ask adjustment and the additional 0.5 bp charge, risk reversal, bear put spread, and put ratio spread retain positive reproduced mean PNL, while the remaining strategy families are negative on average. The corresponding Sharpe ratios also decline. This comparison shows that mid-quote estimates are an optimistic benchmark and that statistical conclusions about unconditional profitability are sensitive to the definition of the response variable used for implementation.

3.4 Tail-risk characteristics

Strategy	Skewness	ES1%	Max DD	Worst Day	Worst 5-Day	Loss Prob (%)	Obs
Strangle/Straddle	1.474	1.423	19.546	-2.511	-7.094	69.255	953
Iron Butterfly/Condor	-0.416	0.579	15.513	-0.658	-1.648	45.750	953
Risk Reversal	3.058	1.601	7.230	-2.766	-4.389	41.868	953
Bull Call Spread	0.372	0.904	20.364	-0.976	-3.947	55.404	953
Call Ratio Spread	-0.285	1.209	22.753	-2.515	-4.410	54.040	953

Strategy	Skewness	ES1%	Max DD	Worst Day	Worst 5-Day	Loss Prob (%)	Obs
Bear Put Spread	0.670	0.827	8.209	-0.864	-2.968	61.765	952
Put Ratio Spread	0.677	0.710	6.111	-0.873	-2.550	58.824	952

Table 4. Reproduced tail-risk diagnostics for implementable net PNL.

The tail-risk estimates reinforce the inadequacy of mean-variance summaries alone. The reproduced risk reversal retains positive average implementable PNL but has a 1% expected shortfall of approximately 1.601% of the underlying and a worst one-day outcome of approximately -2.766%. The call ratio spread has negative skewness and the largest reproduced maximum drawdown, approximately 22.753%. These magnitudes are large relative to average daily PNL and demonstrate that favorable first-moment estimates can coexist with substantial downside exposure.

3.5 Reproduction relative to the original paper

The supplied reproduction is qualitatively consistent with the paper's central Part III message: unconditional strategy performance is heterogeneous, temporally unstable, sensitive to execution costs, and characterized by material tail risk. Exact numerical equality is not obtained for every table. A principal reason is sample coverage. The reproduced implementability and tail-risk tables contain approximately 952-953 daily observations for most strategies, whereas the latest paper reports approximately 1,318-1,319 observations. Differences in data vintage, upstream cleaning, interpolation, aggregation, and cost construction can therefore propagate into means, Sharpe ratios, skewness, expected shortfall, and drawdown.

Strategy	Statistic	Reproduced	Original paper	Difference (Reproduced - Original)
Strangle/Straddle	Net mean B/A + 0.5 bp	-0.0110	-0.0097	-0.0013
Risk Reversal	Net mean B/A + 0.5 bp	0.0106	0.0090	0.0016
Bear Put Spread	Net mean B/A + 0.5 bp	0.0085	0.0022	0.0063
Put Ratio Spread	Net mean B/A + 0.5 bp	0.0198	0.0115	0.0083
Risk Reversal	Tail skewness	3.0580	5.5500	-2.4920
Risk Reversal	ES 1%	1.6010	1.8215	-0.2205
Call Ratio Spread	Tail skewness	-0.2850	-2.0500	1.7650
Put Ratio Spread	Maximum drawdown	6.1110	9.0600	-2.9490

Table 5. Selected statistical differences between the supplied reproduction and Vilkov (2026). The difference is defined as reproduced minus original-paper value; values are shown on the same scale used in the corresponding source tables.

Table 5 makes the numerical reproducibility gap explicit. For implementable mean PNL, the reproduced strangle/straddle estimate is slightly more negative than the paper benchmark, whereas the reproduced risk-reversal, bear-put-spread, and put-ratio-spread means are higher. Tail statistics show larger sensitivity: reproduced risk-reversal skewness is substantially below the paper value, its ES1% is lower, and the reproduced put-ratio-spread maximum drawdown is smaller. These differences are descriptive comparisons rather than formal tests of equality, because the two sets of statistics are computed from different effective samples and the sampling covariance between the estimates is unavailable.

These discrepancies are treated as part of the reproducibility analysis rather than as errors to be hidden. Reproduced quantities are reported as generated from the supplied computational package; values reported only in the paper are treated as benchmark evidence. The same principle is applied to the author's alternative-entry-time and subperiod robustness analyses: they are relevant to interpretation but are not presented as independently reproduced unless they are generated by the supplied project code.

4. Limitations

The first limitation concerns the processed nature of the data. Because raw Cboe SPXW quotes are not reconstructed, the project cannot independently validate the quote filters, interpolation procedure, or all

upstream variable definitions. The estimands in this report are therefore conditional on the supplied processed sample. This is a reproducibility study of the downstream statistical analysis, not a fully independent replication of the original data-generating pipeline.

Second, the effective sample used by the supplied reproduction differs from the latest version of the paper for several analyses. This sample-vintage mismatch is particularly important for statistics that depend strongly on rare observations. Expected shortfall at the 1% level, sample skewness, extrema, and maximum drawdown can change substantially when the sample is extended by several hundred trading days. Numerical differences in these quantities should therefore not be interpreted automatically as computational mistakes.

Third, the observations exhibit dependence. Multiple configurations within the same trading day share common market shocks, while daily financial outcomes may also display serial dependence and volatility clustering. Date-clustered standard errors address within-date dependence in the structural-break regression, but the descriptive estimators in the static tables do not constitute a complete time-series model. Moreover, the 63-observation rolling means are highly overlapping and should not be treated as independent estimates.

Fourth, the analysis involves multiple strategy families and moneyiness configurations. Examining many empirical means, Sharpe ratios, and tail summaries increases the possibility that apparently attractive configurations arise partly through multiple comparison. The project therefore treats these quantities as descriptive reproduction evidence rather than as confirmatory discovery of profitable rules. A stronger inferential claim would require a pre-specified testing framework, multiplicity control, or independent out-of-sample validation.

Fifth, the heavy-tailed nature of ODTE PNL limits the stability of moment- and tail-based estimators. The Sharpe ratio can be sensitive to extreme observations, and ES1% is estimated from only the worst fraction of the sample. The empirical tail measures are consequently useful diagnostics of realized downside risk but should not be interpreted as precisely known population constants.

Finally, implementability is approximated rather than observed directly. Half-spread costs and the additional 0.5 bp slippage-and-fee assumption provide a transparent stress adjustment, but realized trading costs depend on liquidity, order type, execution timing, and market conditions. The structural-break model is also deliberately narrow: it tests a change in conditional mean PNL, not equality of the full pre- and post-period distributions. These limitations restrict the strength of the conclusions, but they do not undermine the main statistical finding of the reproduction: unconditional ODTE performance must be evaluated as a distributional, time-varying, and cost-sensitive phenomenon rather than by average PNL alone.

Source basis. Vilkov, G. (2026), “ODTE Trading Rules,” March 18, 2026 version; supplied processed datasets, unconditional-analysis code, Jupyter notebook, and reproduced figures and tables.